

CHAPTER 9

Architecture Capability



This chapter is part of the GREEN and GOLD reading tours. See Figure 1-1 in Chapter 1, Section 1.3, for alternative reading tours.

This chapter deals with the concept of the Architecture Capability. It refers to an organization's ability to develop, maintain, and govern its Enterprise Architecture effectively. The Architecture Capability is crucial for achieving successful business transformation and maintaining architectural governance.

The TOGAF Standard sees the Architecture Capability as a structured approach to establishing and operating an Enterprise Architecture function within an organization. It includes key components such as architecture governance, organizational structures, processes, roles, responsibilities, and skills necessary to support the architecture practice. The capability helps organizations ensure that architectural efforts are standardized, repeatable, and aligned with corporate strategy.

A well-developed Architecture Capability enables an organization to respond effectively to change, manage complexity, and improve decision-making by providing a structured approach to Enterprise Architecture management. It ensures that architecture-related initiatives are properly governed, resourced, and aligned with business priorities.

9.1. The Ability to Practice Architecture

In order to establish and maintain an architecture in an organization, the creation of an appropriate capability is a must. The TOGAF Standard calls this capability the Architecture Capability. It is about an organization's *ability to practice architecture*.

A loose definition for a capability is:

The ability of an organization to do something.

The word *something* in the definition above should obviously be replaced with the specific subject for which the capability is intended or created – in this case, *to practice architecture*.

9.2. Creating the Architecture Capability

According to the TOGAF Standard, creating the Architecture Capability is no different from creating any other capability. For this reason, it is similar to creating a sustainable architecture. Therefore, the framework indicates that the creation of the Architecture Capability can be supported by going through the phases of the Architecture Development Method.

As with any business capability, the establishment of an Enterprise Architecture Capability can be supported by the TOGAF Architecture Development Method [11].

The TOGAF Standard views the Architecture Development Method as the ideal means to design and oversee the implementation of the Architecture Capability. Applying the Architecture Development Method, particularly Phase A (Architecture Vision), ensures the capability's establishment within an organization.

This capability should not be viewed as a separate phase of an architecture project nor as a stand-alone project. Rather, it should be viewed as an ongoing discipline that provides the context, environment, and resources to govern and enable the delivery of the architecture to the organization.

Implementing any capability requires designing the four architecture domains: Business, Data, Application, and Technology. Therefore, implementing the Architecture Capability follows a similar pattern to developing an average architecture, using the Architecture Development Method.

Table 9-1 shows the phases of the Architecture Development Method and the associated focus areas that must be traversed and addressed to create the Architecture Capability.

Table 9-1. *The Phases and focus areas of the ADM*

ADM Phase	Focus Areas
Phase A: Architecture Vision	Define the project, identify stakeholder concerns and business requirements, identify business goals and drivers, define scope, define constraints, perform Architecture Maturity Assessment.
Phase B: Business Architecture	Agree on a common language, define the process, identify views and viewpoints, decide on the framework, define performance metrics, agree on the governance framework.
Phases C and D: Data, Application, and Technology Architectures	Specify and govern the structure of the Architecture Repository, define requirements for architecture deliverables, define technology.

(continued)

Table 9-1. *(continued)*

ADM Phase	Focus Areas
Phase E: Opportunities and Solutions	Determine organizational change.
Phases F and G: Migration Planning and Implementation Governance	Govern organizational change.
Phase H: Architecture Change Management	Manage change.
Requirements Management	Manage requirements.

The creation of the Architecture Capability begins with *Phase A (Architecture Vision)*:

- **Define the Project:** This step focuses on defining the stakeholders. Stakeholders should include the roles and organizational units involved in the Architecture Capability, as well as the people who will benefit from the deliverables of the Architecture Capability. This step largely determines the positioning of the Architecture Capability in the organization. Therefore, it is a very important step.
- **Identify Stakeholder Concerns and Business Requirements:** This step generates the initial, very high-level definitions of the Architecture Capability from a business information systems and technology perspective.
- **Identify Business Goals and Drivers:** Understanding the business goals and drivers is essential to aligning the Architecture Capability with the organization.

Much like the first step, this step partially determines the positioning of the Architecture Capability in the organization. The exact positioning of the capability depends largely on the business goals being pursued.

- **Define Scope:** Defining the scope of the Architecture Capability cannot be omitted. It cannot be implemented without a high-level project plan of what architecture needs to be addressed in the coming period.
- **Define Constraints:** This step focuses on the organization-wide constraints that affect all architecture projects, including the Architecture Capability (which isn't a one-time project, as mentioned earlier).
- **Perform Architecture Maturity Assessment:** It is essential to conduct an initial Architecture Maturity Assessment to gain insight into the maturity of the current view of architecture within the organization. The way in which the Architecture Capability can be implemented will largely depend on the maturity level of the organization.

Moving through *Phase B (Business Architecture)*, the focus shifts to the following core components:

- **Agree on a Common Language:** Define the architectural terms and definitions that will be used in and by the organization. This will enable the use of a common language that everyone involved in the Architecture Capability will understand.

- **Define the Process:** The Architecture Development Method forms the basis for the architectural process to be used. When using the TOGAF Standard, it is essential to adapt the Architecture Development Method to the context for which it is being used. Use only those parts of the method that contribute to the creation of Architecture Capability.
- **Identify Views and Viewpoints:** Work with stakeholders to determine which views and viewpoints are important to flesh out. One viewpoint that should not be overlooked is architecture governance.
- **Decide on the Framework:** Agree on the creation of architecture deliverables and the requirements for creating these deliverables.
- **Define Performance Metrics:** Establish realistic metrics against which to measure the performance and progress of creating and governing the Architecture Capability.
- **Agree on the Governance Framework:** Formulate what the Governance Framework will look like. Draw a diagram of the framework and document it along with the other agreements.

Phases C and D (Information Systems Architectures, and Technology Architectures) perform the following activities related to the creation of the Architecture Capability:

- **Specify and Govern the Structure of the Architecture Repository:** Design or develop the metamodel to be used with the Architecture Capability. This metamodel describes the entities used by the Architecture Capability, their characteristics, and the key relationships between these entities.

- **Define the Requirements for the Architecture**
Deliverables: Describe the requirements for being able to create, maintain, publish, distribute, and govern the architecture deliverables. A focus here is the selection of an appropriate toolset for creating and maintaining models and diagrams. The ability to publish - making the architecture artifacts findable and available - is also an important focus.
- **Define Technology:** Define the technology and infrastructure that will be used to create and operate the Architecture Capability.

Phase E (Opportunities and Solutions) focuses on determining the organizational change required to accommodate the Architecture Capability:

- **Determine Organizational Change:** Determine the organizational change that is necessary to establish and implement the Architecture Capability and how to achieve this.

The focus of *phases F and G (Migration Planning and Implementation Governance)* is to govern the implementation of the defined steps:

- **Govern Organizational Change:** The Migration Planning phase focuses on implementing the steps defined in the Business Architecture phase. This also applies to Phase G (Implementation Governance).

The Architecture Development Method cycle ends with *Phase H (Architecture Change Management)*:

- **Manage Change:** In this final phase, the focus should be on managing the changes required to establish and implement the Architecture Capability.

During the *Requirements Management Phase*, all requirements should be controlled:

- **Manage Requirements:** All requirements defined during the previous phases should be clearly articulated and aligned. This can be done by continuously following the Requirements Management Phase.

The approach described here makes it all sound and look very formal.

In practice, of course, all the steps mentioned in the Architecture Development Method are performed, but in a less deliberate way. Often, the steps and focus areas described above are reviewed and agreed upon during one or two meetings.

It is recommended that the agreements made are recorded and, if possible, formally signed off by the main sponsor of the Architecture Capability.

The applicable business goals and desired outcomes are determined during the creation of the Architecture Capability. This largely determines the Architecture Capability's position in the organization.

9.3. Positioning the Architecture Capability

Organizations that understand that Enterprise Architecture is a strategic business management tool, not a technical instrument, have a great advantage in setting the direction of the organization. When viewed as a strategic tool, architecture is directly involved in and used to translate organizational strategy into concrete implementation. This scenario is

consistent with the use of the Architecture Capability as described in the Leader's Guide [3]. This guide describes *four ways* to use the Architecture Capability, which are visualized in Figure 9-1.

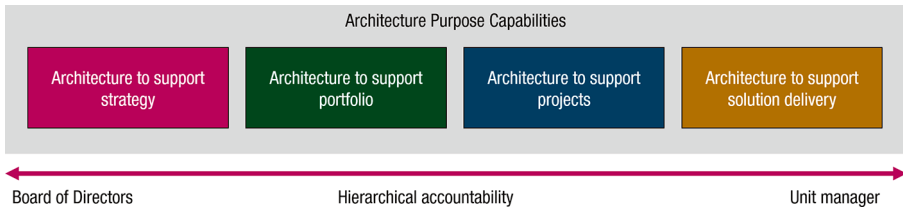


Figure 9-1. *Architecture Purpose Capabilities*

The four Architecture Capabilities shown in Figure 9-1 can be explained as follows:

Architecture to Support Strategy: The Architecture Capability is used to develop target architectures and create roadmaps that reflect organizational changes within the organization over a three- to ten-year period.

In this context, the Architecture Capability is used to support organization-wide change requests and multiple projects and programs and to achieve strategy execution through these projects and programs. There is ample room for the Architecture Capability to shape strategic aspects such as drivers, goals, objectives, and initiatives. Architecture used to support strategy is architecture in its most comprehensive form. The Architecture Capability would benefit greatly from being led by a CEO.

Architecture to Support Portfolio: The Architecture Capability supports change initiatives that span multiple functions, phases, and projects.

An Architecture Capability used for this purpose will primarily focus on a single portfolio. An example of a single portfolio might be migrating to a cloud environment. Within the portfolio, which has a clearly defined scope, multiple projects will require the support of the Architecture Capability. In this context, architecture is used to interpret and guide projects. The architect ensures the alignment of the projects within the

portfolio and the coherence of the projects and steers their execution. The Architecture Capability is thus less of an organizational steering function than when it is primarily concerned with strategy execution.

Architecture to Support Projects: The Architecture Capability is tailored to the project delivery method of the organization.

An Architecture Capability used for this purpose typically covers one project. To stay in the context of migrating to a cloud environment, consider moving an on-premises customer relationship management system to its counterpart in a cloud environment. This migration requires the architect's involvement not only in the Application Architecture domain, but certainly in the Business and Information Architecture domains as well. In the context of supporting projects, the Architecture Capability is used to clarify the purpose and value of the project, as well as to identify requirements and monitor adherence to architecture governance. In addition, architecture supports integration and alignment across projects.

An Architecture Capability that supports portfolios and/or projects should be led by a CIO or CTO. There is still a strategic component to the architecture work that needs to be done, albeit in a slightly milder form than when the Architecture Capability is led by a CEO.

Architecture to Support Solution Delivery: The Architecture Capability supports the delivery of solutions.

An Architecture Capability used for this purpose typically covers one project or a significant portion of a project. The architecture is used to define how the change will be designed and delivered and to identify constraints, controls, and architectural requirements for the design. The architecture provides a Governance Framework for the change. Enterprise Architecture used to support solution delivery is actually similar to Solution Architecture. If the organization chooses this use of architecture, the question arises as to whether the organization would benefit more from employing a Solution Architect than an Enterprise Architect.

This last use of the Architecture Capability can be led by a unit manager. The responsibility for implementing solutions rests with this particular role.

During the creation of the Architecture Capability, it is important to define and agree on how the organization views and interprets the capability and to align it with the defined business goals and drivers. To determine where the primary focus of the capability should be, an Architecture Maturity Assessment can be performed. Such a maturity assessment helps to determine the current level of maturity of the Architecture Capability in the organization. The assessment will provide a clear indication of how the Architecture Capability should be positioned in the organization to achieve business goals. This is critical because the position of the capability is an indication of how the organization views the capability.

The position of the capability in the organization will also determine the line of reporting for the Architecture Capability. The higher the direct reporting line is in the organizational hierarchy, the more the Architecture Capability moves toward the strategic side of the profession. Figure 9-2 provides some insight into the possible placement of the Architecture Capability within the organization.

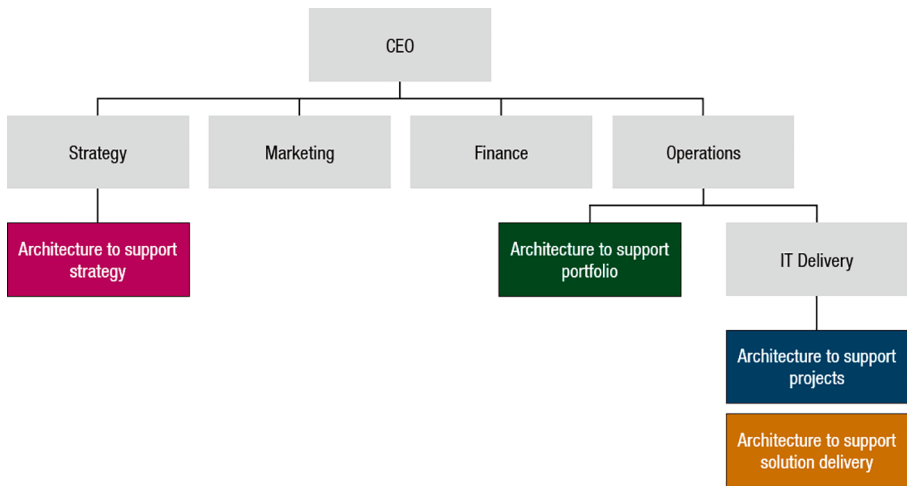


Figure 9-2. *Positioning of the Architecture Capability*

For example, Figure 9-2 shows the four Architecture Capabilities positioned on an organizational chart. The capability that uses architecture to support strategy is often positioned with a direct reporting line to the CEO or to a strategy executive. This ensures that the capability operates primarily in the strategic domain. If the architecture is used to support the portfolio, the direct reporting line will most likely be to the CIO. The architecture used in this scenario will be involved in portfolio and project management. A third way to use the Architecture Capability is to support projects. Using the architecture to structure project execution, as with using it to support the portfolio, will further reduce its scope within the organization. When architecture is used to support projects, the direct reporting line is often to the unit manager of an IT department. Finally, placing the Architecture Capability in the IT domain or department will create the smallest scope for architecture work, as it will focus primarily on IT solution delivery.

Ultimately, there is no right or wrong way to establish an Architecture Capability in an organization. Its position depends entirely on an organization’s needs and desires for working with architecture.

It should be noted, however, that the further to the right the Architecture Capability is positioned - that is, toward the operational side of the organization chart in Figure 9-2 - the more the focus will be on solution-oriented issues. As a result, architecture work will be more reactive and much less proactive. Also, the goals and objectives of an organization will be represented by and in architecture work to a much lesser extent, making architecture less likely to be connected to the strategic direction of the organization. The risk here is that architecture becomes a stand-alone entity that lacks coherence with other business activities.

The Architecture Capability is directly related to the architecture roles distinguished in the TOGAF Standard. These roles are in turn related to the architecture domains.



Further Reading

For more information on the Architecture Capability, see Chapters 5, 9, 11, and 12 of “The TOGAF® Leader’s Guide to Establishing and Evolving an EA Capability”, document number G184.

9.4. Summary

This chapter demonstrated that in order to establish and maintain an architecture in an organization, the creation of an appropriate capability is a must. The TOGAF Standard calls this capability the Architecture Capability.

According to the framework, creating the Architecture Capability is no different from creating any other capability. For this reason, it is similar to creating a sustainable architecture. Therefore, the TOGAF Standard indicates that the creation of the Architecture Capability can be supported by going through the phases of the Architecture Development Method.

Establishing the capability should not be viewed as a separate phase of an architecture project or as a stand-alone project, but rather as an ongoing discipline that provides the context, environment, and resources to govern and enable the delivery of the architecture to the organization.

The creation of the Architecture Capability consists of the following steps:

- **Define the Project:** Focuses on defining the stakeholders.
- **Identify Stakeholder Concerns and Business Requirements:** Generates the initial definitions of the Architecture Capability from a business information systems and technology perspective.
- **Identify Business Goals and Drivers:** This step partially determines the positioning of the Architecture Capability in the organization.
- **Define Scope:** Defines the scope of the Architecture Capability.
- **Define Constraints:** This step focuses on the organization-wide constraints that affect the Architecture Capability.
- **Perform Architecture Maturity Assessment:** The way in which the Architecture Capability can be implemented will largely depend on the maturity level of the organization.
- **Agree on a Common Language:** Define the architectural terms and definitions that will be used in and by the organization.

- **Define the Process:** Use only those parts of the Architecture Development Method that contribute to the creation of the Architecture Capability.
- **Identify Views and Viewpoints:** Work with stakeholders to determine which views and viewpoints are important to flesh out.
- **Decide on the Framework:** Agree on the creation of architecture deliverables and the requirements for creating these deliverables.
- **Define Performance Metrics:** Establish realistic metrics against which to measure the performance and progress of creating and governing the Architecture Capability.
- **Agree on the Governance Framework:** Formulate what the Governance Framework will look like.
- **Specify and Govern the Structure of the Architecture Repository:** Design or develop the metamodel to be used with the Architecture Capability.
- **Define the Requirements for the Architecture Deliverables:** Describe the requirements for being able to create, maintain, publish, distribute, and govern the architecture deliverables.
- **Define Technology:** Define the technology and infrastructure that will be used to create and operate the Architecture Capability.
- **Determine Organizational Change:** Determine the organizational change that is necessary to establish and implement the Architecture Capability and how to achieve this.

- **Govern Organizational Change:** The Migration Planning phase focuses on implementing the Architecture Capability.
- **Manage Change:** In this final phase, the focus should be on managing the changes required to establish and implement the Architecture Capability.
- **Manage Requirements:** All requirements defined during the previous steps should be clearly articulated and aligned.

After discussing the creation and establishment of the Architecture Capability, the chapter showed what its positioning can do for an organization. This positioning is addressed in the *Identify business goals and drivers* step when establishing the Architecture Capability.

There are four possible ways of positioning the capability, each depending on the objectives of the organization:

- **Architecture to Support Strategy:** The Architecture Capability is used to develop target architectures and create roadmaps that reflect organizational changes within the organization over a three- to ten-year period.
- **Architecture to Support Portfolio:** The Architecture Capability supports change initiatives that span multiple functions, phases, and projects.
- **Architecture to Support Projects:** The Architecture Capability is tailored to the project delivery method of the organization.
- **Architecture to Support Solution Delivery:** The Architecture Capability supports the delivery of solutions.

During the creation of the Architecture Capability, it is important to define and agree on how the organization views and interprets the capability and to align it with the defined business goals and drivers. To determine where the primary focus of the capability should be, an Architecture Maturity Assessment can be performed. Such a maturity assessment helps to determine the current level of maturity of the Architecture Capability in the organization.

The position of the capability in the organization will also determine the line of reporting for the Architecture Capability. The higher the direct reporting line is in the organizational hierarchy, the more the Architecture Capability moves toward the strategic side of the profession.